Web Appendix

The Color of Status: Color Saturation, Brand Heritage, and Perceived Status of Luxury Brands

Xinyue Zhou

Chunqu Xiao

Sunyee Yoon

Hong Zhu

Contents

Web Appendix A

Study W1: Marketplace Analysis

Our research proposes that the use of less saturated colors, as opposed to highly saturated ones, can enhance consumer perceptions of brand status when the brand is considered luxury (vs. non luxury). To examine whether luxury brands already employ this strategy commonly compared to non-luxury brands, we analyzed the color saturation in women's bags from three luxury brands and three non-luxury brands.

Method

We have selected three luxury brands and three non-luxury brands. For the luxury brands, we included Louis Vuitton, Chanel, and Gucci, which are top-ranking luxury brands listed on the Interbrand 2022 report (Interbrand 2022). Although Hermès was also on the list, it was not included because most of its bags are not displayed on its website. All three luxury brands are frequently featured in questions assessing consumers' luxury consumption in the marketing literature (e.g., Berger and Ward 2010; Chen, Wang, and Ordabayeva 2023; Wang and Griskvicius 2014; Ward and Dahl 2014).

Three non-luxury brands included Zara and H&M, both ranked as the top two retail brands in the Interbrand 2022 list. Since the Interbrand 2022 retail brand list includes only two fashion brands, we additionally selected a third brand, Banana Republic. The selection of Banana Republic was based on its production of the highest number of women's handbags among the subsidiaries of Gap Inc., one of the biggest clothing retail companies (CSI market 2022). These three non-luxury brands are commonly included in questions evaluating consumers' interest in mass-market brands in the literature (e.g., Miller and Allen 2012; Ordabayeva and Chandon 2011; Sun, Bellezza, and Paharia 2021; Ward and Dahl 2014; Wang, Xu, and Zhang 2023). While other mass-market brands, such as Old Navy, Gap, and American Eagle, are also frequently referenced in the literature, they were excluded from our marketplace study because they rarely offer women's handbags.

A separate study was conducted to ensure that the luxury and non-luxury brands selected were significantly different in terms of luxuriousness but not in terms of popularity. This study was preregistered (https://osf.io/rsfcu). The study involved 300 female participants from Connect; only female consumers were recruited because the focal product was a women's handbag ($M_{\text{age}} = 44.01$). Participants were presented with one of six brands and rated both the luxuriousness ("To what extent do you think this brand is categorized as a luxury brand?" 1 = not at all, 7 = completely) and the popularity of the brand ("People around me recognize this brand," "People around me use this brand," "This brand is popular"; 1 = strongly disagree, 7 = strongly agree; $\alpha = .75$). As expected, the three luxury brands scored significantly higher than the midpoint of 4 ($M_{\text{Gucci}} = 6.41$, SD = .86, $t(48) = 19.51$, $p < .001$; $M_{\text{Chanel}} = 6.40$, SD = .83, $t(49) = 20.37$, $p < .001$; $M_{\text{Louis Vuitton}} = 6.59$, SD = .70, $t(50) = 26.49$, $p < .001$) and did not differ significantly from each other ($F(2, 147) = .89$, $p = .412$). Similarly, the three non-luxury brands scored significantly lower than the midpoint of 4 ($M_{\text{Banana Republic}} = 3.53$, SD = 1.47, $t(50) = -2.28$,

p = .027; $M_{\text{H\&M}}$ = 3.14, SD = 1.44, $t(48)$ = -4.16, p < .001; M_{Zara} = 3.38, SD = 1.43, $t(49)$ = -3.07, p = .003) and showed no significant differences among themselves ($F(2, 147)$ = .90, p = .408). Popularity ratings did not differ significantly across the six brands ($F(5, 294)$ = 1.36, p = .240). This result implies that our selection of six brands might be appropriate for distinguishing between luxury and non-luxury categories based on their perceived luxuriousness, while ensuring that all six brands are widely recognized and popular among consumers.

We downloaded images of all the bags from the official websites of these six brands as of March 17th, 2024. This approach resulted in a total of 1,806 bag images. When excluding black and white bags, we were left with 1,366 images. To expedite calculations, we converted all images to 2,500 pixels and removed the image backgrounds to eliminate the influence of background color. Subsequently, we retrieved color saturation and brightness values from each pixel, ranging from 0% to 100%, based on the HSB model using Python packages (Yu, Xie, and Wen 2020). These values were averaged to form saturation and brightness indices. We did not calculate the hue index as averaging hue values could misconstrue the overall color composition of the image.

Results and Discussion

Saturation. A t-test was conducted to compare the saturation level of luxury and non-luxury products. The result revealed that there was no significant difference (M_{luxury} = 24.18%, SD = .22 vs. $M_{\text{non-luxury}}$ = 23.57%, SD = .21; $t(1804)$ = .51, p = .611, d = .02). We additionally compared the saturation level of luxury and non-luxury products after excluding black and white colored bags for robustness. The result revealed that there was no significant difference (M_{luxury} = 31.74%, SD = .20 vs. $M_{\text{non-luxury}}$ = 29.98%, SD = .20; $t(1364)$ = 1.39, p = .164, d = .08).

Brightness. While the research focuses on saturation, we additionally examined the difference in brightness. The result revealed that there was no significant difference (M_{luxury} = 44.49%, SD = .23 vs. $M_{\text{non-luxury}}$ = 42.97%, SD = .24; $t(1804)$ = 1.18, p = .237, d = .06). After excluding black and white colored bags, we found products from luxury brand are brighter (M_{luxury} = 51.67%, SD = .20 vs. $M_{\text{non-luxury}}$ = 46.98%, SD = .22; $t(1364)$ = 3.59, p < .001, d = .19).

	The number of products	Saturation	Brightness
Luxury brands			
Louis Vuitton	616 (457)	25.83% (34.48%)	44.99% (51.83%)
Chanel	163 (107)	20.56% (30.41%)	34.03% (42.66%)
Gucci	585 (461)	23.45% (29.34%)	46.89% (53.61%)
Non-luxury brands			
ZARA	252 (216)	25.42% (29.24%)	45.82% (46.20%)
H&M	98 (49)	18.99% (36.20%)	38.74% (52.19%)
Banana Republic	92 (76)	23.41% (28.06%)	39.69% (45.82%)

Note. Values in parentheses indicate when black and white colored bags were excluded.

The results suggest that luxury brands, in comparison to their non-luxury counterparts, do not exhibit a clear tendency to use either more or less saturated colors in their products. These findings imply that the use of less saturated colors may not be a strategy more commonly employed by luxury brands relative to non-luxury brands.

Web Appendix B

Study W2: Expert Survey

Study W2 was designed to examine whether decision-makers or fashion designers in the luxury fashion industry commonly utilize less saturated colors to enhance brand heritage or status. We aimed to recruit 50 participants from Prolific and posted a survey link to target individuals who were either currently employed in luxury fashion brands or had prior experience in such brands in roles as fashion designers or managerial positions. After excluding 24 participants who reported having no work experience in the luxury fashion industry, the final sample consisted of 26 participants (61.54% female, $M_{age} = 32.27$). The questions asked and their corresponding responses are presented below:

Question	Response
Screening question (n = 50)	
Do you have work experience in the luxury fashion industry?	• Yes (52 %) • No (48 %)
Means to enhance brand heritage/status (n = 26)	
If you want consumers to perceive your brand as more prestigious and luxurious than competing brands, what kinds of tactics would you use in designing your products? (open-ended)	Coded responses • Mentioning colors (7.7%) • Others (92.3%)
If you want consumers to perceive your brand as more prestigious and luxurious than competing brands, how would you choose the colors of the product? Do you have any specific ideas about what kinds of colors or what aspects of colors can enhance the luxuriousness of your brand? (open-ended)	Coded responses • Mentioning hue (53.8%) • Mentioning brightness (11.5%) • Mentioning saturation (3.8%) • Others (30.8%)
If you want consumers to perceive your brand as having a greater heritage than competing brands, how would you choose the colors of the product? Do you have any specific ideas about what kinds of colors or what aspects of colors can enhance the perceived brand heritage? (open-ended)	Coded responses • Mentioning hue (46.2%) • Mentioning brightness (19.2%) • Mentioning saturation (0%) • Others (34.6%)
The importance of hue, brightness, and saturation (n = 26)	
Colors have three dimensions; hue, brightness, and saturation (vividness). To enhance the luxuriousness of the product, which aspect of color do you think is important? Please rank three dimensions in order of importance.	Rank first: • Saturation (38.5%) • Hue (30.8%) • Brightness (30.8%) Rank second: • Hue (42.3%) • Saturation (42.3%) • Brightness (15.4%)

	Rank third: • Brightness (53.8%) • Hue (26.9%) • Saturation (19.2%)
To enhance perceived brand heritage, which aspect of color do you think is important? Please rank three dimensions in order of importance.	Rank first: • Saturation (46.2%) • Brightness (30.8%) • Hue (23.1%) Rank second: • Hue (46.2%) • Saturation (34.6%) • Brightness (19.2%) Rank third: • Brightness (50.0%) • Hue (30.8%) • Saturation (19.2%)
The effect of color saturation on luxuriousness and heritage (n = 26)	
What is your intuition about the effect of color saturation on the luxuriousness of a brand?	• Highly saturated colors enhance the luxuriousness of a brand (50%). • Less saturated colors enhance the luxuriousness of a brand (30.8%). • There is no clear relationship between color saturation and the luxuriousness of a brand (19.2%).
What is your intuition about the effect of color saturation on perceived brand heritage?	• There is no clear relationship between color saturation and the brand heritage of a brand (38.5 %). • Highly saturated colors enhance the brand heritage of a brand (30.8 %). • Less saturated colors enhance the brand heritage of a brand (30.8 %).
The usefulness of finding (n = 26)	
If you learn from recent research finding that less saturated color can make the product/brand feel more luxurious, do you think this finding will be useful to you?	• Yes (80.8 %) • No (19.2 %)

In sum, this study suggests that there is limited agreement among individuals in the luxury fashion industry regarding how a product's color saturation influences the perceived status and heritage of a brand.

Web Appendix C

Study W3: Implicit Association Test

We conducted an Implicit Association Test (IAT; Greenwald, Nosek, and Banaji 2003) study to assess whether consumers' association between color saturation and the notion of heritage is implicit. This study was preregistered (https://aspredicted.org/7MJ_4BM).

Method

We recruited 150 participants who used desktops or laptops for the survey on MTurk. As preregistered, eighteen participants who responded to less than 80% of the tasks and two participants who did not pass the color blindness test were excluded from the analyses. The final sample comprised 130 adults (53.85% female, M_{age} = 41.80).

Participants were instructed to categorize six higher color-saturated and six less color-saturated product images, as well as four heritage-related words (i.e., heritage, traditional, historical, classic) and four recency-related words (i.e., new, modern, recent, novel).

The task consisted of seven blocks (i.e., three practice blocks and four critical blocks). Congruent critical blocks required participants to use one key to categorize low saturated colors and heritage-related words and another key to categorize high saturated colors and relatively contemporary-related words. Incongruent critical blocks required participants to use the same key to categorize low saturated colors and relatively contemporary-related words and another same key to categorize high saturated colors and heritage-related words. The order of the incongruent and congruent blocks (i.e., blocks 4-7) was counterbalanced.

Participants' reaction time in categorizing stimuli was recorded as the dependent variable. As suggested by Greenwald, Debbie, and Jordan (1998), a reaction time below 0.3 of a second was recorded as 0.3 seconds, and that above 3 seconds was recorded as 3 seconds.

Results and Discussion

We adopted two different methods to analyze the reaction time. First, we analyzed only correct responses (i.e., responses from participants who had followed the instructions correctly) and conducted a t-test. Findings showed that participants reacted significantly faster in the congruent blocks than in the incongruent blocks ($M_{\text{congruent}}$ = 1.16, SD = .29 vs. $M_{\text{incongruent}}$ = 1.23, SD = .27; $t(129) = 4.13$, $p < .001$, d = .73).

Second, a D-score algorithm [($M_{\text{incongruent}}$ - $M_{\text{congruent}}$)/$SD_{\text{congruent}}$] with a 0.6 second penalty for incorrect responses was calculated (Greenwald et al. 2003). The average accuracy rate was 91.3%. Again, a t-test showed that participants reacted faster in the congruent (vs. incongruent) conditions with a 0.6 second penalty for incorrect responses ($M_{\text{congruent}}$ = 1.24, SD = .33 vs. $M_{\text{incongruent}}$ = 1.30, SD = .32; $t(129) = 3.26$, $p = .001$, d = .57). This difference was statistically significant, as the mean D-score was significantly different from zero (M_{d} = .15, SD = .53; $t(129) = 3.25$, $p = .001$, d = .57).

Although twenty-three participants chose "yes" to the system-delay question ("When you were doing the classification task, did you experience system lag or delay [i.e., you could not see the next screen immediately after pressing the button]?"), excluding them did not change the significance of the results. Both the t-test of the reaction time only with correct responses ($M_{congrouent}$ = 1.18, SD = .30 vs. $M_{incongruent}$ = 1.23, SD = .28; $t(106) = 2.74$, $p = .007$, d = .53) and the D-score algorithm with a 0.6 second penalty for incorrect responses (M_d = .10, SD = .53; $t(106) = 2.03$, $p = .045$, d =.39) were significant.

In sum, this study showed that participants made congruent associations (i.e., images with lower color saturation and heritage-related words, images with higher color saturation and recency-related words) more quickly than incongruent associations (i.e., images with lower color saturation and recency-related words, images with higher color saturation and heritage-related words), supporting that consumers might have an implicit association between less saturated colors and the notion of heritage.

Stimuli used in study W3

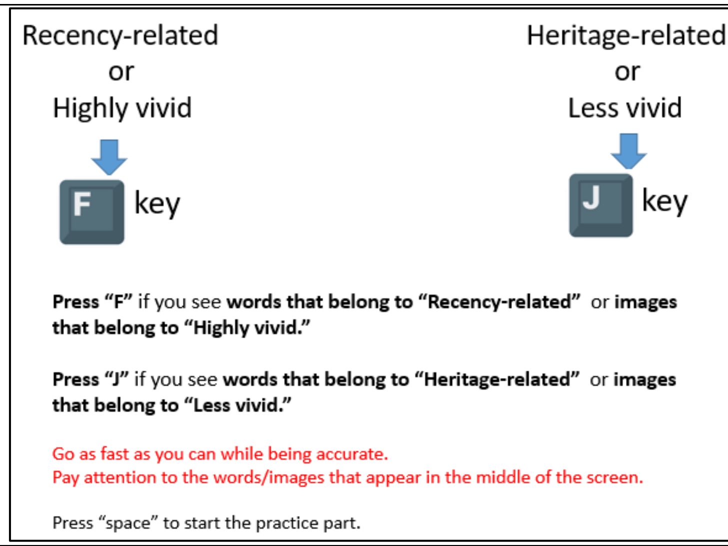
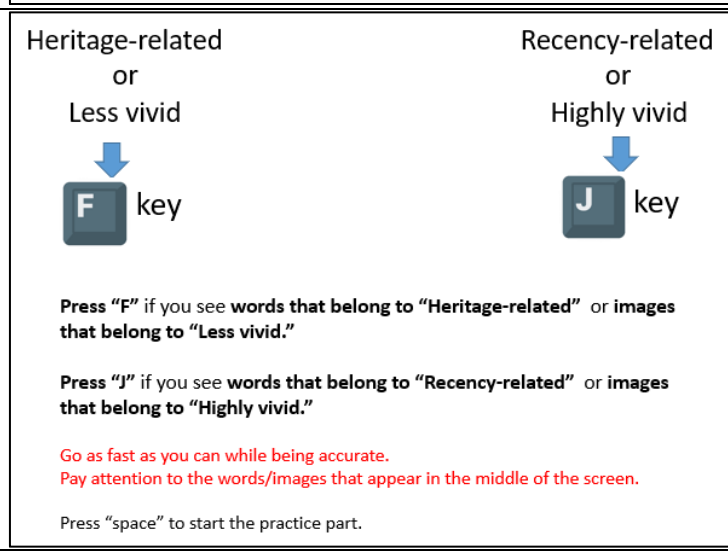
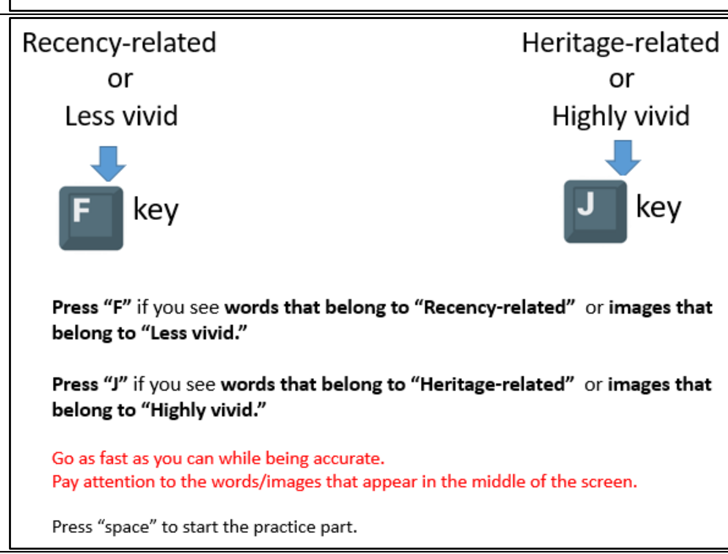
Category	Images
High color saturation	
Low color saturation	

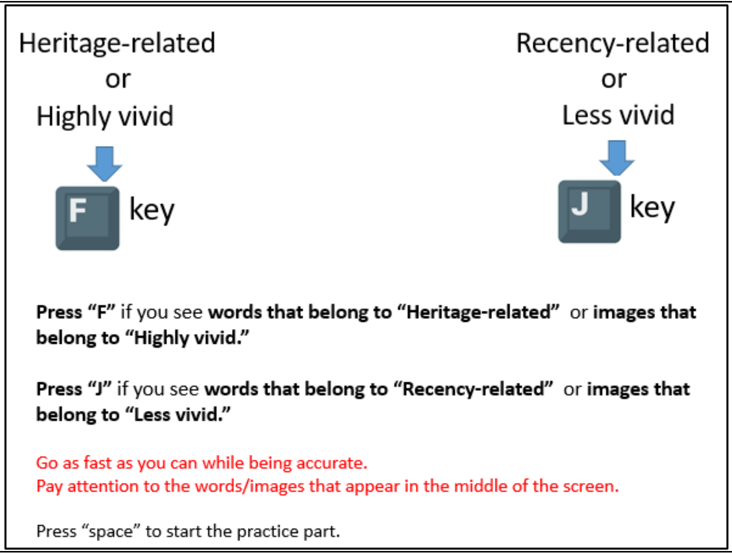
Design of IAT

Block	No. of trials	Stimuli assigned the left-key response	Stimuli assigned the right-key response
1 (practice)	8	Heritage-related word	Recency-related word
2 (practice)	12	Highly vivid color	Less vivid color
3 (practice)	4	Recency-related word or highly vivid color	Heritage-related word or less vivid color
4 (congruent)	20	Recency-related word or highly vivid color	Heritage-related word or less vivid color
5 (congruent)	20	Heritage-related word or less vivid color	Recency-related word or highly vivid color
6 (incongruent)	20	Recency-related word or less vivid color	Heritage-related word or highly vivid color
7 (incongruent)	20	Heritage-related word or highly vivid color	Recency-related word or less vivid color

Instructions for each block

Block	Instruction
Block 1	Heritage-related Recency-related This is an introduction for the first practice part. Please put a left finger on the "F" key and a right finger on the "J" key. You only need to use these two keys and the space bar in the test. **Please don't use the mouse during the task.** As you can see, there are two category words in the top corners of the screen. In the task, you will see another word in the middle of the screen. **Press "F"** if you see words that belong to the category word in the **top left corner (Heritage-related).** **Press "J"** if you see words that belong to the category word in the **top right corner (Recency-related).** Go as fast as you can while being accurate. Pay attention to the words that appear in the middle of the screen. Press "space" to start the practice part.
Block 2	Highly vivid Less vivid This is an introduction for the second practice part. Please put a left finger on the "F" key and a right finger on the "J" key. **Please don't use the mouse during the task.** As you can see, there are two category words in the top corners of the screen. In the task, you will see an image in the middle of the screen. **Press "F"** if you see images that belong to the category word in the **top left corner (Highly vivid).** **Press "J"** if you see images that belong to the category word in the **top right corner (Less vivid).** Go as fast as you can while being accurate. Pay attention to the images that appear in the middle of the screen. Press "space" to start the practice part.
Block 3	Recency-related or Highly vivid — Heritage-related or Less vivid F key — J key **Press "F"** if you see **words that belong to "Recency-related"** or **images that belong to "Highly vivid."** **Press "J"** if you see **words that belong to "Heritage-related"** or **images that belong to "Less vivid."** Go as fast as you can while being accurate. Pay attention to the words/images that appear in the middle of the screen. Press "space" to start the practice part.

<table>
<tr>
<td>Block 4</td>
<td>
</td>
</tr>
<tr>
<td>Block 5</td>
<td>
</td>
</tr>
<tr>
<td>Block 6</td>
<td>
</td>
</tr>
</table>

Block 7	Heritage-related or Highly vivid F key Recency-related or Less vivid J key **Press "F"** if you see **words that belong to "Heritage-related"** or **images that belong to "Highly vivid."** **Press "J"** if you see **words that belong to "Recency-related"** or **images that belong to "Less vivid."** Go as fast as you can while being accurate. Pay attention to the words/images that appear in the middle of the screen. Press "space" to start the practice part.

Web Appendix D

Study W4: Moderation by Brand Type (Luxury vs. Non-luxury)

The objective of study W4 was to replicate the finding of study 2 using different hue and brand type descriptions. We preregistered this study (https://osf.io/wyu76).

Method

Study W4 used a between-subjects design: color saturation (high, low) × brand type (luxury, non-luxury). We recruited 405 participants from Prolific. As preregistered, we excluded five participants failing the color blindness test and two participants reporting image loading issues. The final sample comprised 398 participants (62.31% female, $M_{\text{age}} = 43.46$).

Participants viewed an image of a green watch that was either high or low in color saturation. Participants were told that “This watch is from a luxury fashion brand.” (luxury brand condition) or “This watch is from a fashion brand, which is not considered as luxury” (non-luxury brand condition). Participants reported the perceived brand status as in study 1 ($\alpha = .96$).

Stimuli used in study W4

High saturation condition	Low saturation condition
	

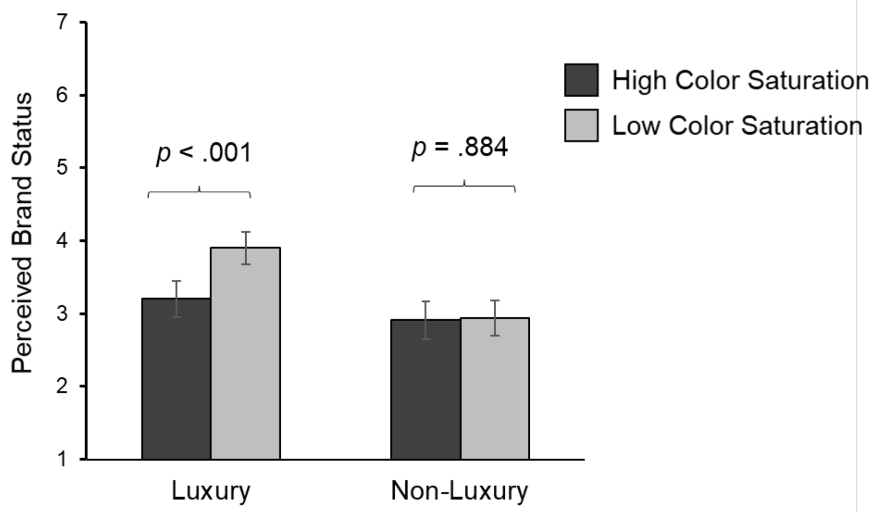
Note: High saturation: saturation 50%, hue 110°, brightness 73%; Low saturation: saturation 19%, hue 110°, brightness 73%

Results and Discussion

A 2 (color saturation) × 2 (brand type) ANOVA was conducted on perceived brand status. It revealed a significant interaction ($F(1, 394) = 6.62$, $p = .010$, partial $\eta^2 = .02$) with significant main effects of color saturation ($F(1, 394) = 7.71$, $p = .006$, partial $\eta^2 = .02$) and brand type ($F(1, 394) = 21.93$, $p < .001$, partial $\eta^2 = .05$). Further analyses revealed low (vs. high) color saturation enhanced perceived brand status in the luxury brand condition ($M_{\text{high saturation}} = 3.19$, SD = 1.29 vs. $M_{\text{low saturation}} = 3.90$, SD = 1.44; $F(1, 394) = 14.10$, $p < .001$, partial $\eta^2 = .04$) but not in the non-luxury brand condition ($M_{\text{high saturation}} = 2.91$, SD = 1.33 vs. $M_{\text{low saturation}} = 2.94$, SD = 1.24; $F(1, 394) = .02$, $p = .884$, partial $\eta^2 < .01$). Also, perceived brand status was rated higher for a luxury (vs. non-luxury) brand when the product was low in color saturation ($F(1, 394) = 26.18$, $p < .001$, partial $\eta^2 = .06$) but not when it was high in color saturation ($F(1, 394) = 2.24$, $p = .136$, partial $\eta^2 = .01$).

Study W4 showed that low (vs. high) color saturation enhanced perceived brand status when the target brand was luxury, but not when the brand was not luxury, replicating study 2.

Figure W1. Interaction of color saturation and brand type



Note. Error bars indicate the 95% confidence interval.

Web Appendix E

Study W5: Role of Continuity Heritage-Saturation Association

Study W5 sought to provide additional support for the mechanism based on continuity heritage. We theorized that low color saturation of a product increases perceived brand status because consumers associate low color saturation with continuity heritage. Then, weakening the association (i.e., a story telling that young brands often use less saturated colors) should mitigate the color saturation effect. We preregistered this study (https://osf.io/cvjah).

Method

Design and Participants. Study W5 used a between-subjects design: color saturation (high, low) × association (strengthened vs. weakened). We recruited 407 people from Prolific. After excluding three participants failing the color blindness test and one participant reporting an image loading issue, the final sample comprised 403 participants (65.26% female, $M_{\text{age}} = 39.18$).

Procedure. Participants read a story that either strengthened or weakened the association between less saturated colors and continuity heritage. Specifically, in the strengthened association condition, they were told,

"Luxury brands with a rich history and heritage often incorporate muted and less vivid colors into their products. This preference for less vivid colors stems from the idea that colors usually lose their vividness over time. Today, muted and less vivid colors have become a symbol of a brand's heritage, and as such, luxury brands often use them in their designs."

In the weakened association condition, they were told,

"Many luxury brands often incorporate muted and less vivid colors into their products. This preference for less vivid colors stems from the idea that colors usually lose their vividness over time. Thus, many designers purposefully use muted and less vivid colors to manipulate a sense of heritage, even when their brand is young and lacks a long-established history or legacy."

Next, participants viewed an image of a bag that was either high or low in color saturation with a description that this bag was from a luxury brand. Participants reported perceived brand status as in study 1 ($\alpha = .96$).

Stimuli used in study W5

High saturation condition	Low saturation condition
	

Note: High saturation: saturation 54%, hue 183°, brightness 75%; Low saturation: saturation 13%, hue 183°, brightness 75%

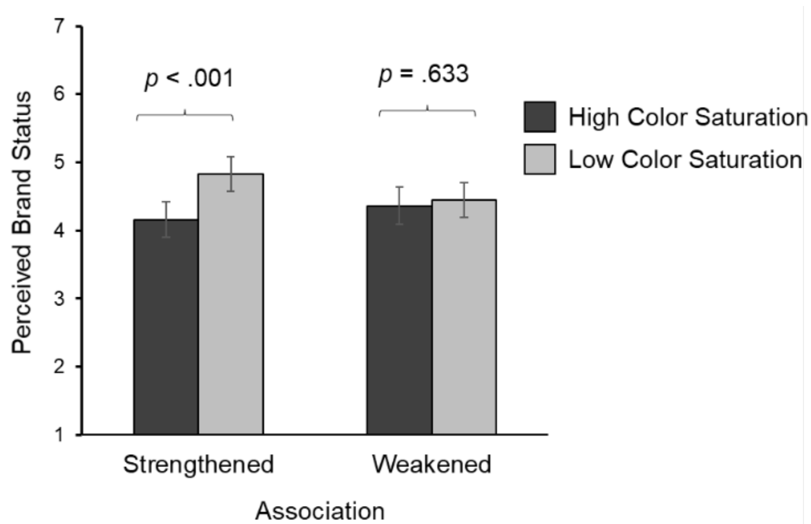
Results and Discussion

A 2 (color saturation) × 2 (association) ANOVA revealed a significant interaction ($F(1, 399) = 4.94$, $p = .027$, partial $\eta^2 = .01$) and a significant main effect of color saturation ($F(1, 399) = 8.39$, $p = .004$, partial $\eta^2 = .02$). The main effect of association was not significant ($F(1, 399) = .41$, $p = .521$, partial $\eta^2 < .01$).

As expected, low (vs. high) color saturation resulted in a greater perceived brand status when the association was strengthened ($M_{\text{high saturation}} = 4.16$, SD = 1.31 vs. $M_{\text{low saturation}} = 4.83$, SD = 1.31; $F(1, 399) = 13.07$, $p < .001$, partial $\eta^2 = .03$), but not when it was weakened ($M_{\text{high saturation}} = 4.36$, SD = 1.39 vs. $M_{\text{low saturation}} = 4.45$, SD = 1.29; $F(1, 399) = .23$, $p = .633$, partial $\eta^2 < .01$). Strengthening (vs. weakening) association increased perceived brand status in the low color saturation condition ($F(1, 399) = 4.11$, p = .043, partial $\eta^2 = .01$), but not in the high color saturation condition ($F(1, 399) = 1.25$, p = .265, partial $\eta^2 < .01$).

Study W5 revealed that when participants learn that less saturated colors do not necessarily signify brand heritage, the effect of color saturation on brand status can be attenuated. This study provided support for the continuity heritage-based mechanism and learned color association.

Figure W2. Interaction of color saturation and association



Note. Error bars indicate the 95% confidence interval.

Web Appendix F

Study W6: Moderation by Luxury Brand Experience

Study W6 was conducted to test another alternative explanation that luxury brands may frequently use less saturated colors in their products, leading consumers to perceive less saturated colors as more luxurious. We expected that the color saturation effect would be stronger for consumers with greater experience with luxury brands if less saturated colors were indeed more common for luxury brands. Study W1 (web appendix A) has already shown that luxury brands, as compared to non-luxury brands, do not produce more products in less saturated colors. Nevertheless, we revisited this possibility by assessing the moderation by consumers' experiences with luxury brands. We preregistered this study (https://osf.io/9eaxp).

Method

Design and Participants. Study W6 used a between-subjects design with two color saturation conditions (high, low). We recruited 399 participants from Credamo. After excluding three participants failing the color blindness test and three participants reporting image loading issues, the final sample comprised 393 adults (63.61% female, $M_{\text{age}} = 39.23$).

Procedure. Participants viewed an image of a watch that was either high or low in color saturation with a description that the watch was a luxury brand. Next, they reported perceived brand status as in study 1 ($\alpha = .89$). Participants' experience with luxury brands was measured using five items ("I am knowledgeable about luxury fashion brands," "I often browse luxury products online or in brick and mortar stores," "I am familiar with many luxury fashion brands," "I am frequently exposed to luxury products," "I can see many luxury products in my daily lives"; 1 = "strongly disagree" to 7 = "strongly agree"; $\alpha = .93$).

Stimuli used in study W6

High saturation condition	Low saturation condition
	

Note: High saturation: saturation 57%, hue 112°, brightness 55%; Low saturation: saturation 17%, hue 112°, brightness 55%

Results and Discussion

Participants in the low (vs. high) color saturation condition perceived higher brand status ($M_{\text{high saturation}}$ = 5.48, SD = 1.18 vs. $M_{\text{low saturation}}$ = 5.83, SD = .98; $t(391) = 3.13$, $p = .002$, d = .32). A multiple regression procedure revealed that when perceived brand status was regressed on color saturation (0 = high, 1 = low) and luxury brand experience, both color saturation (b = .30, SE = .10, $t = 2.88$, $p = .004$) and luxury brand experience (b = .28, SE = .04, $t = 7.59$, $p < .001$) were significant. When we included the interaction between saturation and experience in this model, the interaction was not significant (b = .07, SE = .07, $t = .92$, $p = .361$).

Study W6 showed that consumers' experience with luxury brands does not moderate the color saturation effect. This implies that the effect of color saturation on brand status may not be attributed to the prevalence of less saturated colors in luxury brands.

Web Appendix G

Study W7: WTP and the Moderation by Need for Status Consumption

Study W7 explored the downstream consequences of the color saturation effect, specifically focusing on consumers' willingness to pay (WTP). Consumers may be more willing to pay for a luxury brand's less saturated product because they perceive it as having a higher status. Thus, we tested the effect of color saturation on WTP, with perceived brand status serving as a mediator. We further tested the moderating role of the need for status consumption. Specifically, we predicted that products with low (vs. high) saturation would be perceived as having higher status, leading consumers with a greater need for status consumption to be willing to pay more for these products. In other words, the need for status consumption may moderate the relationship between perceived brand status and WTP.

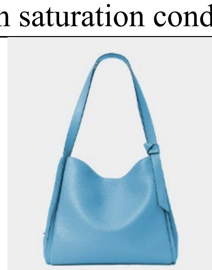
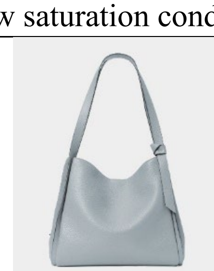
To test these predictions, we conducted a two-wave study. In the first wave, we measured consumers' need for status consumption, and in the second wave, we manipulated color saturation and measured consumers' WTP and perceived brand status. We preregistered this study (https://osf.io/rqus8).

Method

Design and Participants. We recruited 3,000 participants from Credamo in the first wave and randomly selected 400 participants from this pool for the second wave. We collected the first-wave data from September 26th to 27th, 2024, and the second-wave data on October 5th, 2024. After excluding one participant who failed the color blindness test and two participants who reported an image loading issue in the second wave, the final sample consisted of 397 adults (66.50% female, M_{age} = 30.37).

Procedure. In the first wave, participants reported their need for status consumption (Eastman, Ronald, and Reinecke 1999; e.g., "I would buy a product just because it has status"; 1 = "strongly disagree," 7 = "strongly agree"; α = .91). In the second wave, the study employed a between-subjects design with two color saturation conditions (high and low). Participants viewed an image of a bag that was either high or low in color saturation with a description that the product was a luxury brand. Participants then reported their willingness to pay for the product ("Please enter the amount that you are willing to pay for this product in CNY, ranging from 0 to 20,000 CNY") and rated perceived brand status as in study 1 (α = .93).

Stimuli used in study W7

High saturation condition	Low saturation condition
	

Note: High saturation: saturation 50%, hue 201°, brightness 78%; Low saturation: saturation 12%, hue 201°, brightness 78%

Results and Discussion

As expected, participants in the low (vs. high) color saturation condition reported higher WTP ($M_{\text{high saturation}}$ = 5598.35, SD = 5611.54 vs. $M_{\text{low saturation}}$ = 6784.22, SD = 5711.09; $t(395)$ = 2.09, p = .038, d = .21), suggesting that a product's less saturated colors increase consumers' WTP. Next, a bootstrapping analysis (PROCESS Model 4, with 5,000 resamples; Hayes 2017) was conducted to examine if perceived brand status mediates the effect of color saturation on WTP. Color saturation was entered as the independent variable (0 = high, 1 = low), WTP as the dependent variable (higher values indicate higher willingness to pay), and perceived brand status as the mediator (higher values indicate greater perceived brand status). When perceived brand status was regressed on color saturation, color saturation predicted perceived brand status ($M_{\text{high saturation}}$ = 4.87, SD = 1.34 vs. $M_{\text{low saturation}}$ = 5.38, SD = 1.06; b = .51, SE = .12, t = 4.22, p < .001). When WTP was regressed on both color saturation and perceived brand status, perceived brand status significantly predicted WTP (b = 2433.76, SE = 201.68, t = 12.07, p < .001), but color saturation was nonsignificant (b = -60.15, SE = 497.04, t = -.12, p = .904). The indirect effect of perceived brand status was significant (b = 1246.02, SE = 308.85, $CI_{95\%}$ [658.20, 1855.66]), supporting the effect of color saturation on WTP and the mediation by perceived brand status.

To test the moderation by the need for status consumption, we conducted a bootstrapping analysis (PROCESS Model 15, with 5,000 resamples; Hayes 2017). In this analysis, we specified color saturation as the independent variable (0 = high, 1 = low), WTP as the dependent variable, perceived brand status as the mediator, and need for status consumption as the moderator between the mediator and dependent variable. We applied PROCESS Model 15, expecting that the need for status consumption moderates the relationship between perceived brand status and WTP. When perceived brand status was regressed on color saturation, color saturation significantly predicted perceived brand status (b = .51, SE = .12, t = 4.22, p < .001). Next, WTP was regressed on color saturation, perceived brand status, need for status consumption, and the interactions between need for status consumption and each of the other variables. The interaction between perceived brand status and need for status consumption was significant (b = 370.65, SE = 125.74, t = 2.95, p = .003), while perceived brand status (b = 665.92, SE = 578.09, t = 1.15, p = .250), color saturation (b = -537.17, SE = 1642.76, t = -.33, p = .744), need for status consumption (b = -1281.32, SE = 666.66, t = -1.92, p = .055), and the interaction between color saturation and need for status consumption (b = 134.08, SE = 345.73, t = .39, p = .698) were nonsignificant. Importantly, the moderated mediation index of perceived brand status was significant (index = 189.76, SE = 79.15, $CI_{95\%}$ [62.02, 375.13]), supporting the moderating role of need for status consumption.

In sum, this study demonstrated that the color saturation effect extends to consumers' willingness to pay. Specifically, products with low (vs. high) saturation are perceived as having higher status, leading consumers with a greater need for status consumption to be willing to pay more for these products.

Web Appendix H

Study W8: Perceived Consumer Status

Study W8 examined whether a product's color saturation affects the perceived status of product users. We expected that a person using a luxury product with low (vs high) color saturation would be considered to have a higher social status. We preregistered this study (https://osf.io/vk25f).

Method

Design and Participants. Study W8 used a between-subjects design with two color saturation conditions (high, low). We recruited 200 participants from Credamo. After excluding one participant failing the color blindness test and three participants reporting an image loading issue, the final sample consisted of 196 adults (69.90% female, $M_{\text{age}} = 32.42$).

Procedure. Participants were asked to imagine that they were attending a high-end banquet and noticed a woman who might have a high status. Specifically, they were told,

"Imagine attending a high-end banquet, a prime opportunity for networking and connecting with new people. Your goal is to find a high-status individual who can offer a partnership opportunity to your company. As you wander around, wine glass in hand, you consider who among the attendees might have such status. You notice a woman holding a bag, and you observe that the bag looks like this."

Please note that we did not directly mention in the scenario that the bag is from a luxury brand. The scenario was written from the perspective of an observer, and it is quite challenging to notice the brand of the bag from this perspective. Instead, we mentioned that this is happening at a high-end banquet to lead participants to think that attendees might carry luxurious items.

They were then shown an image of the bag with high or low color saturation as below and rated the perceived status of this woman (Cannon and Rucker 2019; "To what extent do you think this person has the following traits: high status/ prestigious / elite / upper class / prominent"; 1 = "not at all," 7 = "very much", $\alpha = .94$).

Stimuli used in study W8

High saturation condition	Low saturation condition
	

Note: High saturation: saturation 55%, hue 38°, brightness 83%; Low saturation: saturation 12%, hue 38°, brightness 83%

Results and Discussion

Participants evaluated that the individual holding a bag with low (vs. high) color saturation had a higher status ($M_{\text{high saturation}}$ = 4.56, SD = 1.44 vs. $M_{\text{low saturation}}$ = 5.08, SD = 1.23; t (194) = 2.71, p = .007, d = .39). Consistent with our expectation, using low (vs. high) color saturation enhances users' perceived brand status. In turn, the color saturation effect extends to consumers' perceived status. To further support this conclusion, we conducted a field observation study which examines whether customers carrying bags with lower color saturation are treated with greater respect compared to those with higher saturation (i.e., Study W9).

Web Appendix I

Study W9: Field Observation Study

Study W9 investigated another consequence of color saturation by observing how consumers carrying bags with high or low color saturation are treated by staff at luxury stores. Luxury stores were chosen as the setting for this field study because the color saturation effect has been pronounced in the luxury domain, as seen in study 2. Moreover, luxury stores are known for the differential treatment of customers based on perceived status (Angela 2024), making the interactions between staff and customers a valuable source of insight for assessing judgments of customer status. Additionally, many luxury stores have implemented strategies to manage crowd flow, resulting in customers waiting in queues outside (Fickenscher 2022). This setup creates easily observable interactions between customers and door hosts, making it an ideal setting for examining the impact of color saturation on customer treatment in real life. We preregistered this study: (https://osf.io/m59gw).

Method

Two raters, who were unaware of the research hypotheses, were recruited from a local university. To minimize the likelihood of the raters inferring the study's purpose, we selected college freshmen who had not taken any courses in consumer behavior or marketing research. The raters were trained to assess color saturation by learning principles of color dimensions and evaluating images of luxury bags. They were asked to categorize each image as having either high or low color saturation until they achieved a 99% accuracy rate.

This field study was conducted over two days, on March 16 and 24, 2024, at Louis Vuitton and Gucci stores in a luxury shopping mall. These locations were chosen because they offered clear, unobstructed views of the interaction between door host and customers. As preregistered, the raters documented 116 valid interactions (58 observations from Louis Vuitton and 58 from Gucci).

Upon arriving at the observation location outside a store entrance, the raters first observed the door hosts' interactions with ten customers to establish a baseline of typical respect levels shown by the hosts in this store. After establishing this baseline, they began the rating task, focusing solely on customers with colored bags and disregarding those carrying black, white, or grey bags. Additionally, raters did not rate interactions when multiple individuals entered the store simultaneously or when the interaction between customers and door hosts was not clearly visible to either of the raters. When a customer carrying a colored bag approached the entrance, the raters quickly reached a consensus to select this customer for observation and independently evaluated the color saturation of the bag and the door host's attitude. Specifically, they categorized the color saturation as either low or high and rated the respect level as below average, average, or above average. They also recorded the approximate age and gender of both customers and door hosts. After completing the observations, the raters were asked to guess the study's purpose; neither accurately identified the hypothesis.

Results and Discussion

Ratings from the two raters demonstrated high intercoder reliabilities ($r_{color\ saturation}$ = 1.00, $p < .001$; $r_{\text{respect}} = .70$, $p < .001$), allowing us to average their ratings. We regressed the averaged respect level (-1 = below the average level; 0 = the average level; 1 = above the average level) on color saturation (0 = high, 1 = low). The result is presented in the table below.

Summary of regression results in Study W9

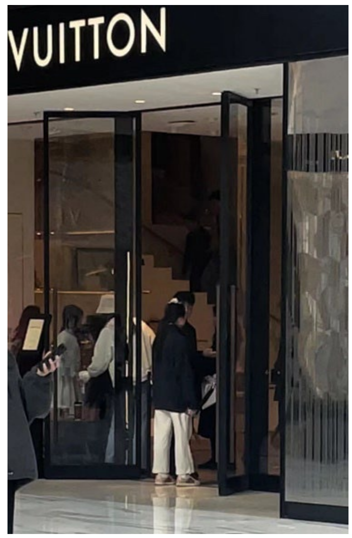
	DV = Respect shown by door hosts			
	Step 1		Step 2	
	b	SE	b	SE
(Constant)	.22**	.09	.35	.92
Predictor				
Color saturation (0 = high, 1 = low)	.37**	.14	.37**	.15
Control Variables				
Brand (0 = Louis Vuitton, 1 = Gucci)			.24	.15
Gender of Customers (0 = female, 1 = male)			-.34*	.19
Gender of Door hosts (0 = female, 1 = male)			.03	.32
Age of Customers (continuous)			-.01	.01
Age of Door hosts (continuous)			<.01	.02
Recorded Date (0 = March 16, 1 = March 24)			-.36	.23
R^2	.05		.15	
ΔR^2	.05**		.09*	
F	6.44**		2.64**	
df	114		108	

NOTE. — Unstandardized coefficients were reported. All variance inflation factors were below 1.25 and tolerance values were above 0.8. *$p < .1$, **$p < .05$, and ***$p < .01$.
Each rater rated the respect shown by door hosts on three levels (-1 = below average level; 0 = average level; 1 = above average level), and these ratings were averaged.

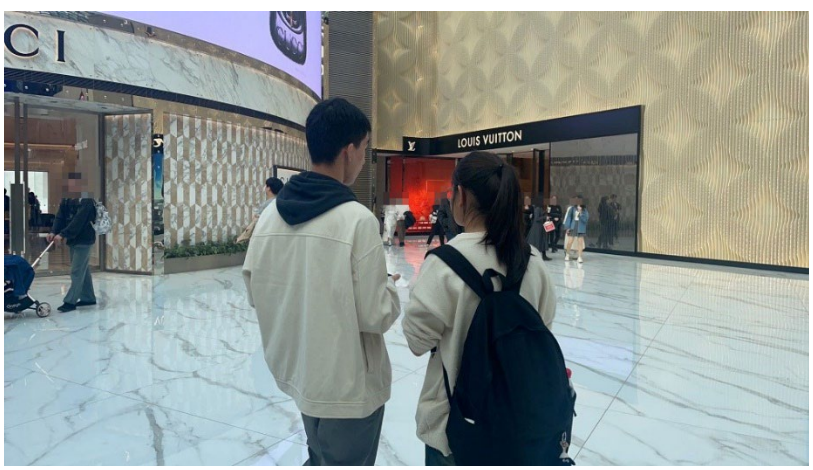
Consistent with our expectation, customers with bags of low (vs. high) color saturation received more respect from door hosts (b = .37, SE = .14, $t = 2.54$, $p = .012$). This effect remained significant (b = .37, SE = .15, $t = 2.55$, $p = .012$), even after controlling for stores (Louis Vuitton = 0, Gucci = 1), the approximate age and gender (0 = female, 1= male) of customer and door hosts, and the date (0 = March 16, 1= March 24).

This study showed that color saturation can influence real-world outcomes by affecting how customers are treated by door hosts at luxury stores. Our field observations revealed that customers carrying bags with lower color saturation are treated with greater respect compared to those with higher saturation. Given that luxury store staff tend to show more respect to consumers who appear to have higher social status (Angela 2024), our results suggest that door hosts perceive customers with less saturated bags as having higher status, thereby treating them with greater respect.

The view of the two stores from the raters’ recording spot



The view of the Gucci store (left) and the Louis Vuitton store (right)



Raters

Web Appendix J

Study 1

J.1. Cell size
(between-subject design, after excluding participants who did not pass the exclusion questions)

Condition	90%	80%	70%	60%	50%	40%	30%	20%	10%
Cell size	99	98	99	99	100	100	99	100	99

J.2. Questionnaire

You will see a watch from a luxury brand.

90%	80%	70%	60%	50%	40%	30%	20%	10%
								

Note: The stimuli's saturation levels range from 90% to 10% in 10% intervals, as shown in the table above. Other color dimensions (hue: 110°, brightness: 73%) remain constant.

Perceived brand status:
Please rate how you feel about the brand. (1 = strongly disagree, 7 = strongly agree)

- The brand has a high status.
- The brand is well-respected.
- The brand is prestigious.
- The brand is high-end.

Age: What is your age (in years)? ___________

Gender: What is your gender?

- Male
- Female

Exclusion question 1: Were you able to see all images clearly in this study?

- Yes
- No

Exclusion question 2: Please write the number in the picture above. If you are color blind, you will not be able to see this number. In this case, you can write "I am color blind." You will be still compensated for your submission.

J.3. Pairwise comparison

	10%	20%	30%	40%	50%	60%	70%	80%
20%	.905							
30%	<.01	<.01						
40%	<.001	<.01	.431					
50%	<.01	<.01	.888	.516				
60%	<.05	<.05	.685	.232	.584			
70%	<.001	<.001	.284	.775	.351	.140		
80%	<.001	<.001	.257	.725	.318	.124	.947	
90%	<.001	<.001	<.05	.093	<.05	<.01	.164	.187

Web Appendix K

Study 2

K.1. Cell size
(between-subject design, after excluding participants who did not pass the exclusion questions)

Condition	Luxury		Non-luxury	
	High saturation	Low saturation	High saturation	Low saturation
Cell size	99	96	99	101

K.2. Questionnaire

- Luxury condition: You will see a product. This product is from a luxury brand.
- Non-luxury condition: You will see a product.

High saturation condition	Low saturation condition
	

Note: High saturation: saturation 56%, hue 340°, brightness 80%; Low saturation: saturation 18%, hue 340°, brightness 80%

Perceived brand status:
Compared with other brands, please rate how you feel about the brand.
(1 = strongly disagree, 7 = strongly agree)

- The brand has a high status.
- The brand is well-respected.
- The brand is prestigious.
- The brand is high-end.

Age: What is your age (in years)? ___________

Gender: What is your gender?

- Male
- Female

Exclusion question 1: Were you able to see all images clearly in this study?

- Yes
- No

Exclusion question 2: Please write the number in the picture above. If you are color blind, you will not be able to see this number. In this case, you can write "I am color blind." You will be still compensated for your submission.

Web Appendix L

Study 3

L.1. Cell size
(between-subject design, after excluding participants who did not pass the exclusion questions)

Condition	High saturation	Low saturation
Cell size	100	97

L.2. Questionnaire

You will see a watch from a luxury brand.

High saturation condition	Low saturation condition
	

Note: High saturation: saturation 50%, hue 110°, brightness 73%; Low saturation: saturation 19%, hue 110°, brightness 73%

Perceived brand status:
Please rate how you feel about the brand. (1 = strongly disagree, 7 = strongly agree)

- The brand has a high status.
- The brand is well-respected.
- The brand is prestigious.
- The brand is high-end.

Perceived continuity heritage:
Please rate how you feel about the brand. (1 = strongly disagree, 7 = strongly agree)

- The brand seems like a brand that survives times.
- The brand seems like a brand with a history.
- The brand seems like a brand that survives trends.
- The brand seems like a timeless brand.

Perceived integrity heritage:
Please rate how you feel about the brand. (1 = strongly disagree, 7 = strongly agree)

- The brand seems like a brand that is true to a set of moral values.
- The brand seems like a brand that cares about consumers.
- The brand seems like a brand with moral principles.

- The brand seems like a brand that gives back to its consumers.

Visual prominence:
What's your perception of this watch? Please rate the extent to which each descriptor describes the watch. (1 = not at all, 7 = very much)

- Showy
- Flashy
- Subtle (reversed)
- Modest (reversed)

Arousal:
What's your perception of this watch? Please rate the extent to which each descriptor describes the watch.

- 1= sluggish 7 = frenzied
- 1= calm 7 = excited
- 1= relaxed 7 = stimulated

Note: The order of continuity heritage, integrity heritage, arousal, and visual prominence was randomized.

Age: What is your age (in years)? __________

Gender: What is your gender?

- Male
- Female

Exclusion question 1: Were you able to see all images clearly in this study?

- Yes
- No

Exclusion question 2: Please write the number in the picture above. If you are color blind, you will not be able to see this number. In this case, you can write "I am color blind." You will be still compensated for your submission.

L.3. Correlation of measured constructs

	Perceived Brand Status	Continuity Heritage	Integrity Heritage	Arousal
Continuity Heritage	.66**			
Integrity Heritage	.45**	.62**		
Arousal	.02	-.19**	-.06	
Visual Prominence	.06	-.18*	-.16*	.54**

NOTE. *p < .05, **p < .01

L.4. Discriminant validity of perceived brand status and continuity heritage

Although the correlation between perceived brand status and continuity heritage was .66, which falls below .85, the upper limit for correlations between two different constructs (Kline 2005), we additionally examined the discriminant validity of these two constructs as below.

<u>Exploratory factor analysis</u>

Rotated sums of squared loadings

Factor	Eigenvalue	% of Variance	Cumulative %
1	3.49	43.66	43.66
2	3.33	41.58	85.24

Rotated factor matrix

Items	1	2
The brand has a high status.	.89	.33
The brand is well-respected.	.84	.40
The brand is prestigious.	.89	.32
The brand is high-end.	.89	.30
The brand seems like a brand that survives times.	.32	.85
The brand seems like a brand with a history.	.33	.86
The brand seems like a brand that survives trends.	.30	.84
The brand seems like a timeless brand.	.33	.84

Extraction Method: Principal Component Analysis.
Rotation Method: Varimax with Kaiser Normalization.

Confirmatory factor analysis

Coefficient Table for the factors

Variable	Item	Coef.	SE	*Z*	*p*	Std. Estimate	CR	AVE
Perceived Brand Status	The brand has a high status.	1.000				.927	.96	.85
	The brand is well-respected.	.929	.043	21.480	<.001	.901		
	The brand is prestigious.	1.000	.043	23.509	<.001	.928		
	The brand is high-end.	1.063	.046	23.062	<.001	.922		
Perceived Continuity Heritage	The brand seems like a brand that survives times.	1.000				.894	.93	.76
	The brand seems like a brand with a history.	1.166	.060	19.309	<.001	.914		
	The brand seems like a brand that survives trends.	.975	.062	15.633	<.001	.824		
	The brand seems like a timeless brand.	1.126	.068	16.575	<.001	.849		

Estimation Method: Maximum Likelihood
χ^2 = 96.28, df = 19, CFI = .95, TLI = .93, NFI = .94, MSV = .48

Fornell-Larker criterion

We compared the square root of the average variance extracted (AVE) with the correlation of two constructs. The square root of each construct's AVE exceeds the correlation (*r* = .66), indicating the discriminant validity of the two constructs (Fornell and Larcker 1981).

Factor	AVE	The square root of AVE
Perceived Brand Status	.85	.92
Perceived Continuity Heritage	.76	.87

Heterotrait-Monotrait ratio of correlations

We employed the HTMT (Heterotrait-Monotrait Ratio of Correlations) method as wll. HTMT values close to 1 indicate a lack of discriminant validity, and the threshold of .90 has been suggested (Gold, Malhotra, and Segars 2001). Given the HTMT value of the two constructs is 0.71, we concluded that these two scales measure different constructs.

L.5. Mediation (detailed analyses)

Mediation by perceived integrity heritage. A bootstrapping analysis (PROCESS Model 4, with 5,000 resamples; Hayes 2017) was conducted with color saturation (0 = high, 1 = low) as an independent variable, perceived brand status (higher values indicate greater brand status) as a dependent variable, and perceived integrity heritage (higher values indicate greater integrity heritage) as a mediator. Color saturation didn't predict perceived integrity heritage (b = .10, SE = .16, t = .62, p = .533). When perceived brand status was regressed on color saturation and perceived integrity heritage, both perceived integrity heritage (b = .51, SE = .07, t = 6.93, p < .001) and color saturation (b = .47, SE = .17, t = 2.83, p = .005) were significant. The indirect effect was not significant (b = .05, SE = .09, $CI_{95\%}$ [-.11, .22]).

Mediation by visual prominence. A bootstrapping analysis (PROCESS Model 4, with 5,000 resamples; Hayes 2017) was conducted with color saturation (0 = high, 1 = low) as an independent variable, perceived brand status (higher values indicate greater brand status) as a dependent variable, and visual prominence (higher values indicate greater visual prominence) as a mediator. Color saturation predicted visual prominence (b = -.67, SE = .18, t = -3.75, p < .001), implying that high (vs. low) color saturation is more visually prominent. When perceived brand status was regressed on color saturation and visual prominence, color saturation was significant (b = .61, SE = .19, t = 3.15, p = .002), while visual prominence was not significant (b = .12, SE = .07, t = 1.59, p = .113). The indirect effect was not significant (b = -.08, SE = .05, $CI_{95\%}$ [-.22, .01]).

Mediation analysis of arousal. A bootstrapping analysis (PROCESS Model 4, with 5,000 resamples; Hayes 2017) was conducted with color saturation (0 = high, 1 = low) as an independent variable, perceived brand status (higher values indicate greater brand status) as a dependent variable, and arousal (higher values indicate greater arousal) as a mediator. Color saturation predicted arousal (b = -.78, SE = .16, t = -4.89, p < .001), implying that high (vs. low) color saturation is more arousing. When perceived brand status was regressed on color saturation and arousal, color saturation was significant (b = .61, SE = .20, t = 3.08, p = .002), while arousal was not significant (b = .10, SE = .08, t = 1.24, p = .215). The indirect effect was not significant (b = -.08, SE = .07, $CI_{95\%}$ [-.25, .05]).

Mediation by perceived continuity heritage, perceived integrity heritage, visual prominence, and arousal. Bootstrapping analysis (PROCESS Model 4, with 5,000 resamples; Hayes 2017) was conducted with color saturation (0 = high, 1 = low) as an independent variable and perceived brand status (higher values indicate greater brand status) as a dependent variable. Perceived continuity heritage, perceived integrity heritage, visual prominence, and arousal were entered as mediators altogether.

As reported earlier, color saturation significantly predicted perceived continuity heritage (b = .52, SE = .19, t = 2.77, p = .006), visual prominence (b = -.67, SE = .18, t = -3.75, p < .001), and arousal (b = -.78, SE = .16, t = -4.89, p < .001), but it did not predict perceived integrity heritage (b = .10, SE = .16, t = .62, p = .533). When perceived brand status was regressed on color saturation, perceived continuity heritage and all three other alternative constructs, color saturation (b = .39, SE = .15, t = 2.67, p = .008), perceived continuity heritage (b = .63, SE = .07,

$t = 9.40$, $p < .001$), and visual prominence (b = .17, SE = .06, $t = 2.61$, $p = .010$) were significant, but perceived integrity heritage (b = .09, SE = .08, $t = 1.21$, $p = .228$) was not significant. Arousal (b = .12, SE = .07, $t = 1.68$, $p = .094$) was marginally significant. The indirect effect of perceived continuity heritage (b = .33, SE = .13, $CI_{95\%}$ [.09, .58]) and visual prominence (b = -.11, SE = .06, $CI_{95\%}$ [-.24, -.02]) were significant. The indirect effects of perceived integrity heritage (b = .01, SE = .02, $CI_{95\%}$ [-.01, .09]) and arousal (b = -.09, SE = .07, $CI_{95\%}$ [-.25, .02]) were not significant.

Web Appendix M

Study 4

M.1. Cell size
(between-subject design, after excluding participants who did not pass the exclusion questions)

Condition	Control		Old foundation year		Recent foundation year	
	High saturation	Low saturation	High saturation	Low saturation	High saturation	Low saturation
Cell size	100	96	97	102	96	98

M.2. Questionnaire
On the next screen, you will be presented with a product from a luxury brand.
Please see the product and answer the following question.

	High saturation	Low saturation
Control		
Old foundation year		
Recent foundation year		

Note: High saturation: saturation 56%, hue 340°, brightness 80%; Low saturation: saturation 18%, hue 340°, brightness 80%
Perceived brand status:
Please rate how you feel about the brand. (1 = strongly disagree, 7 = strongly agree)

- The brand has a high status.
- The brand is well-respected.
- The brand is prestigious.
- The brand is high-end.

Perceived continuity heritage:
Please rate how you feel about the brand. (1 = strongly disagree, 7 = strongly agree)

- The brand seems like a brand that survives times.
- The brand seems like a brand with a history.
- The brand seems like a brand that survives trends.
- The brand seems like a timeless brand.

Visual prominence:
Please rate how you feel about the brand (1 = strongly disagree, 7 = strongly agree)

- Showy
- Flashy
- Subtle (reversed)
- Modest (reversed)

Age: What is your age (in years)? ___________

Gender: What is your gender?

- Male
- Female
- Others

Exclusion question 1: Were you able to see all images clearly in this study?

- Yes
- No

Exclusion question 2: Please write the number in the picture above.
If you are colorblind, please write that you are colorblind.

Web Appendix N

Study 5

N.1. Cell size
(between-subject design, after excluding participants who did not pass the exclusion questions)

Condition	Innovation		Control	
	High saturation	Low saturation	High saturation	Low saturation
Cell size	97	96	99	97

N.2. Questionnaire

You will see a product from a luxury brand.

- Innovation condition:
  This is an excerpt from the webpage: Why settle for the status quo when you can innovate to excel? Embark on a luxurious journey of innovation.

- Control condition:
  This is an excerpt from the webpage: Embark on a luxurious journey.

High saturation condition	Low saturation condition
	

Note: High saturation: saturation 53%, hue 197°, brightness 80%; Low saturation: saturation 13%, hue 197°, brightness 80%

Perceived brand status:
Please rate how you feel about the brand. (1 = strongly disagree, 7 = strongly agree)
- The brand has a high status.
- The brand is well-respected.
- The brand is prestigious.
- The brand is high-end.

Perceived continuity heritage:
The product color makes me think that… (1 = strongly disagree, 7 = strongly agree)
- The brand seems like a brand that survives times.
- The brand seems like a brand with a history.
- The brand seems like a brand that survives trends.
- The brand seems like a timeless brand.

Visual prominence:
The color of the product is… (1 = strongly disagree, 7 = strongly agree)

- Showy
- Flashy
- Subtle (reversed)
- Modest (reversed)

Age: What is your age (in years)? __________

Gender: What is your gender?

- Male
- Female

Exclusion question 1: Were you able to see all images clearly in this study?

- Yes
- No

Exclusion question 2: Please write the number in the picture above. If you are color blind, you will not be able to see this number. In this case, you can write "I am color blind." You will be still compensated for your submission.

Web Appendix O

Study 6

O.1. Cell size
(between-subject design)

Condition	High saturation	Low saturation
Cell size	121	121

O.2. Questionnaire

Page 1:
You will see a luxury bag.
This bag is from a luxury brand sold in this store and is priced at 7,500 yuan. The bag is never discounted.
Your task is to decide the highest price you would be willing to pay for this bag.

Here is how the process works:
A random price X (between 0 and 7,500 yuan) will be assigned to the bag.
One participant from this survey will be randomly selected to receive the prize.

Depending on your bid price for the bag, you will receive different rewards:

- If your bid price is greater than or equal to the random price X:
  You will receive the bag and 7,500 - X yuan in cash.
- If your bid price is lower than the random price X:
  You will NOT receive the bag, but you will receive 7,500 yuan in cash.

Key points to consider:
If you really want the bag, you may want to bid higher.
If you are not very interested in the bag, you can bid lower.
You will see this bag on the next page.

Page 2:

High saturation condition	Low saturation condition
	

Note: High saturation: saturation 55%, hue 208°, brightness 68%; Low saturation: saturation 13%, hue 208°, brightness 68%

Small leather handbag
¥7,500

- With one main compartment
- Adjustable strap
- Magnetic buckle closes
- Dust bag included

Please indicate the highest price you are willing to pay for this bag, ranging from 0 - 7,500 yuan (numeric value only)： __________

Page 3:

Please rate how you feel about the brand. (1 = strongly disagree, 7 = strongly agree)

- The brand has a high status.
- The brand is well-respected.
- The brand is prestigious.
- The brand is high-end.

Page 4:

Age: What is your age (in years)? __________

Gender: What is your gender?

- Male
- Female

Contact information: Your phone number or E-mail: __________

O.3. Pictures Taken in the Field

Poster banner introducing this event

Questionnaires used in the filed

O.4. Results with Log-Transformed WTP

Participants in the low saturation condition exhibited significantly higher log-transformed WTP compared to those in the high saturation condition ($M_{\text{high saturation}}$ = 3.25, SD = .46; $M_{\text{low saturation}}$ = 3.40, SD = .36; t (240) = 2.98, p = .003, d = .38).

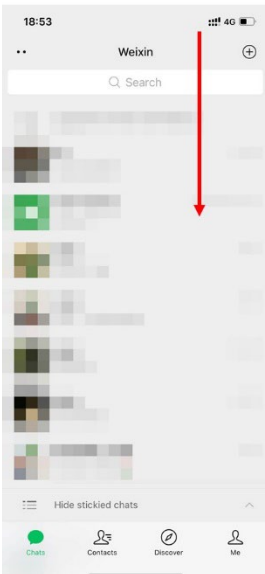
To test the mediation hypothesis, we conducted a bootstrapping analysis using PROCESS Model 4 (Hayes 2017) with 5,000 resamples. Color saturation was entered as the independent variable (0 = high, 1 = low), perceived brand status as the mediator, and log-transformed WTP as the dependent variable. The results revealed that color saturation significantly predicted perceived brand status ($M_{\text{high saturation}}$ = 3.69, SD = 1.12; $M_{\text{low saturation}}$ = 4.02, SD = 1.08; b = .33, SE = .14, t = 2.37, p = .019). When both color saturation and perceived brand status were entered as predictors of log-transformed WTP, perceived brand status significantly predicted log-transformed WTP (b = .20, SE = .02, t = 9.56, p < .001). The effect of color saturation on WTP was significant (b = .09, SE = .05, t = 2.01, p = .046). Most importantly, the indirect effect of color saturation on log-transformed WTP through perceived brand status was significant (b = .07, SE = .03, 95% CI [.01, .13]).

Web Appendix P

Study 7

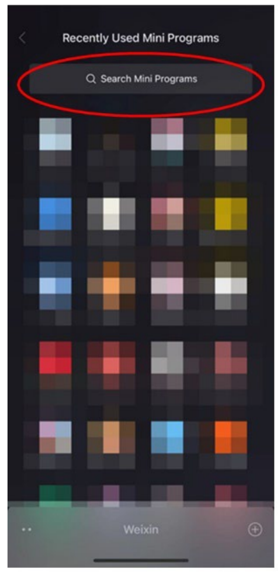
P.1. How to play the game

1.



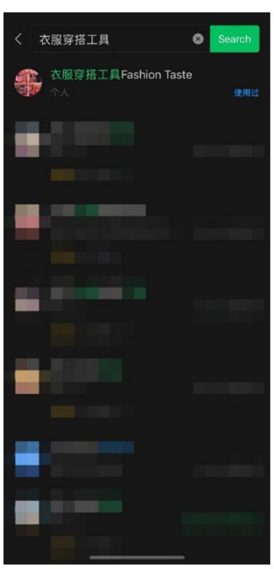
Open the WeChat app. Drop down this interface(see the red arrow)

2.



Search for the key words (衣服穿搭工具) of this game in the search bar in red circle.

3.



The game will be shown in the search results.
Users can click the game and play it in WeChat immediately. They don't need to download this game.

P.2. Details of the game

Captured screen images	Translation
	Fashion Taste Mini-Challenge Tips for privacy protection Thank you for playing the game. Please read the agreement below. <Agreement for privacy protection> By clicking 'Agree and Proceed,' you acknowledge and accept the terms of our Privacy Agreement, which will be legally binding. Alternatively, you can click 'Disagree and Exit' to leave the game. Disagree and exit Agree and proceed
	Tap on your profile avatar to start playing the game!

<table>
<tr>
<td>


</td>
<td>
Introduction

Personal style is often demonstrated through the ability to skillfully select outfits, even with limited options. Meet Xiao Fang, who struggles with choosing the right things to wear. She often finds herself pondering, ‘What should I wear today?’

Imagine you are Xiao Fang. Are you able to select the most suitable outfits for various occasions?

Click ‘I Understand’ to continue
</td>
</tr>
<tr>
<td>


</td>
<td>
How to Play the Game

Below are the instructions on how to play the game:

Access the Closet: Click here to open your closet.
Select Your Items: Choose your top, bottom, and bag.
Switch Categories: Click here to switch between different categories of clothing and accessories.
Choose Products: Simply click on the items to select them.

Click ‘I Understand’ to proceed
</td>
</tr>
<tr>
<td>


</td>
<td>
Scenario 1

You are invited to attend a high-end banquet that elite members attend. You need to make yourself look like one of them.
This is your first challenge. Please choose the most appropriate clothes for this banquet.

Choose Outfits
</td>
</tr>
</table>

<table>
<tr>
<td>
</td>
<td>Choose the clothes from the wardrobe!

Click here to open the wardrobe.</td>
</tr>
<tr>
<td>
</td>
<td>Click here to close the wardrobe

Top Bottom Bags

Submit your choice</td>
</tr>
</table>

	Click here to close the wardrobe Top Bottom Bags Submit your choice
	Click here to close the wardrobe Top Bottom Bags The bags are from luxury brands Submit your choice
	Scenario 2 You are going shopping at a local shopping center. This is your second challenge. Please choose the most appropriate clothes for going shopping. Choose outfits (Note: After Scenario 2, players performed the same tasks while viewing the same options for tops, bottoms, and luxury bags.)

	Your score is: … (Note: the score is a random number between 80 and 90.) If you enjoyed this game, please share it with your friends! "Thank you for choosing the outfits!" Here are your choices; let's review them! Banquet Shopping

P.3. Stimuli

High saturation		Low saturation	
	Saturation: 56% Hue: 226° Brightness: 70%		Saturation: 18% Hue: 226° Brightness: 70%
	Saturation: 56% Hue: 43° Brightness: 80%		Saturation: 15% Hue: 43° Brightness: 80%

	 Saturation: 54% Hue: 0° Brightness: 82%		 Saturation: 15% Hue: 0° Brightness: 82%

Two Sets of Bag Stimuli	
Set A	Set B
	

References

Angela, G.W. (2024), "When Luxury Brands Stop Respecting Their Customers," Last Accessed March 25, 2024. https://medium.com/@angelagw/when-luxury-brands-stop-respecting-their-customers-9c7fae09701c#.

Berger, Jonah and Morgan Ward (2010), "Subtle Signals of Inconspicuous Consumption," *Journal of Consumer Research*, 37 (4), 555-69.

Cannon, Christopher and Derek Rucker (2019), "The Dark Side of Luxury: Social Costs of Luxury Consumption," *Personality and Social Psychology Bulletin*, 45 (5), 767-79.

Chen, Qihui, Yajin Wang, and Nailya Ordabayeva (2023), "The Mate Screening Motive: How Women Use Luxury Consumption to Signal to Men," *Journal of Consumer Research*, 50(2), 303-21.

CSI Market (2022), "GPS's vs. Market Share Relative to its Competitors, as of Q1 2023," Last Accessed August 11, 2023. https://csimarket.com/stocks/competitionSEG2.php?code=GPS.

Eastman, Jacqueline, Goldsmith Ronald, and Flynn Reinecke (1999), "Status Consumption in Consumer Behavior: Scale Development and Validation," *Journal of Marketing Theory and Practice*, 7(3), 41-52.

Fickenscher, Lisa (2022), "Luxury Stores Still Limiting Crowds Post-COVID — And Won't Admit Why," Last Accessed March 25, 2024. https://nypost.com/2022/05/29/luxury-stores-still-limiting-crowds-post-covid-and-wont-admit-why/.

Fornell, Claes and David F. Larcker (1981), "Structural Equation Models with Unobservable Variables and Measurement Error: Algebra and Statistics," *Journal of Marketing Research*, 18(3), 382-8.

Gold, Andrew H., Arvind Malhotra, and Albert H. Segars (2001), "Knowledge Management: An Organizational Capabilities Perspective," *Journal of Management Information Systems*, 18(1), 185-214.

Greenwald, Anthony, McGhee Debbie, and Schwartz Jordan (1998), "Measuring Individual Differences in Implicit Cognition: the Implicit Association Test," *Journal of Personality and Social Psychology*, 74 (6), 1464-80.

Greenwald, Anthony, Brian Nosek, and Mahzarin Banaji (2003), "Understanding and Using the Implicit Association Test: I. An Improved Scoring Algorithm," *Journal of Personality and Social Psychology*, 85 (2), 197-216.

Hayes, Andrew F. (2017), *Introduction to Mediation, Moderation, and Conditional Process Analysis: A Regression-Based Approach*. NY: Guilford Publications.

Interbrand (2022), "Best Global brands," Last Accessed October 15, 2022. https://ibgstaging.wpengine.com/ best-global-brands/.

Kline, Rex B. (2005), *Principles and Practice of Structural Equation Modeling*, NY: Guilford Press.

Miller, Felicia and Chris Allen (2012), "How Does Celebrity Meaning Transfer? Investigating the Process of Meaning Transfer with Celebrity Affiliates and Mature Brands," *Journal of Consumer Psychology*, 22(3), 443-52.

Ordabayeva, Nailya and Pierre Chandon (2011), "Getting Ahead of the Joneses: When Equality Increases Conspicuous Consumption among Bottom-Tier Consumers," *Journal of Consumer Research*, 38(1), 27-41.

Sun, Jennifer, Silvia Bellezza, and Neeru Paharia (2021), "Buy Less, Buy Luxury: Understanding and Overcoming Product Durability Neglect for Sustainable Consumption," *Journal of Marketing*, 85(3), 28-43.

Wang, Yajin and Vladas Griskevicius (2014), "Conspicuous Consumption, Relationships, and Rivals: Women's Luxury Products as Signals to Other Women," *Journal of Consumer Research*, 40(5), 834-54.

Wang, Yajin, Alison Jing Xu, and Ying Zhang (2023), "L'Art Pour l'Art: Experiencing Art Reduces the Desire for Luxury Goods," *Journal of Consumer Research*, 49(5), 786–810.

Ward, Morgan and Darren W. Dahl (2014), "Should the Devil Sell Prada? Retail Rejection Increases Aspiring Consumers' Desire for the Brand," *Journal of Consumer Research*, 41(3), 590-609.

Yu, Joanne, Selina Xie, and Jun Wen (2020), "Coloring the Destination: The Role of Color Psychology on Instagram," *Tourism Management*, 80, 104110.